

where SOAs below some threshold cannot be recovered, so that an observer can only guess about order.¹⁹ However, either kind of model can easily be fitted and interpreted from either theoretical perspective.

8 Choosing between Observer Models and Rejecting Participants

Two further reasonable questions one might ask are: 1) could my observer model have generated these data? and 2) does another observer model describe the data better? Model comparison is a large and complex topic, so once again, what I have to say here should be treated as a brief introduction rather than a comprehensive summary.

Let's begin by considering a metric I have not yet mentioned: *Deviance*. Deviance (sometimes called G^2) is a measure based on log likelihood, but which looks rather more like summed squared error, in that it is zero for a perfectly fitting model and large/positive for a poorly fitting model. Formally, deviance is two times the difference in log likelihood between the *saturated* model and the model with our current set of parameters. A saturated model is one that exactly predicts the data (which can always be accomplished by a model that has one parameter per data point). Hence it represents the situation with the maximum possible log-likelihood when predicting this particular set of data. Deviance is closely related to a simpler calculation ($-2 \times \log \text{likelihood}$) that forms the basis of a couple of well-known metrics for model comparison (the Akaike information criterion, AIC, and the Bayesian information criterion, BIC) and indeed is occasionally defined this way. That's because we are often only really interested in differences (in Deviance, or AIC, or BIC) between models, and the log-likelihood of the saturated model gets subtracted out in a comparison between two models (because it has contributed to the deviance in the same way for both) so calculating it is not necessary.

However, if you want to say something about the goodness of fit of a model *without* relating it to any other model, based on asymptotic statistical theory, you do need to calculate deviance properly. Asymptotically, it turns out that the deviance of a model fitted to data *when that model actually generated those data* follows a chi-square (χ^2) distribution, with degrees of freedom equal to the number of data points minus the number of model parameters (note: for

19 García-Pérez and Alcalá-Quintana's commitment to this account is a little unclear, because they often let δ vary across experimental conditions, suggesting flexibility more akin to a criterion-based account. It may be that they believe a low-threshold exists, but that synchrony is often additionally reported beyond this hard limit.